New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

Part 1:

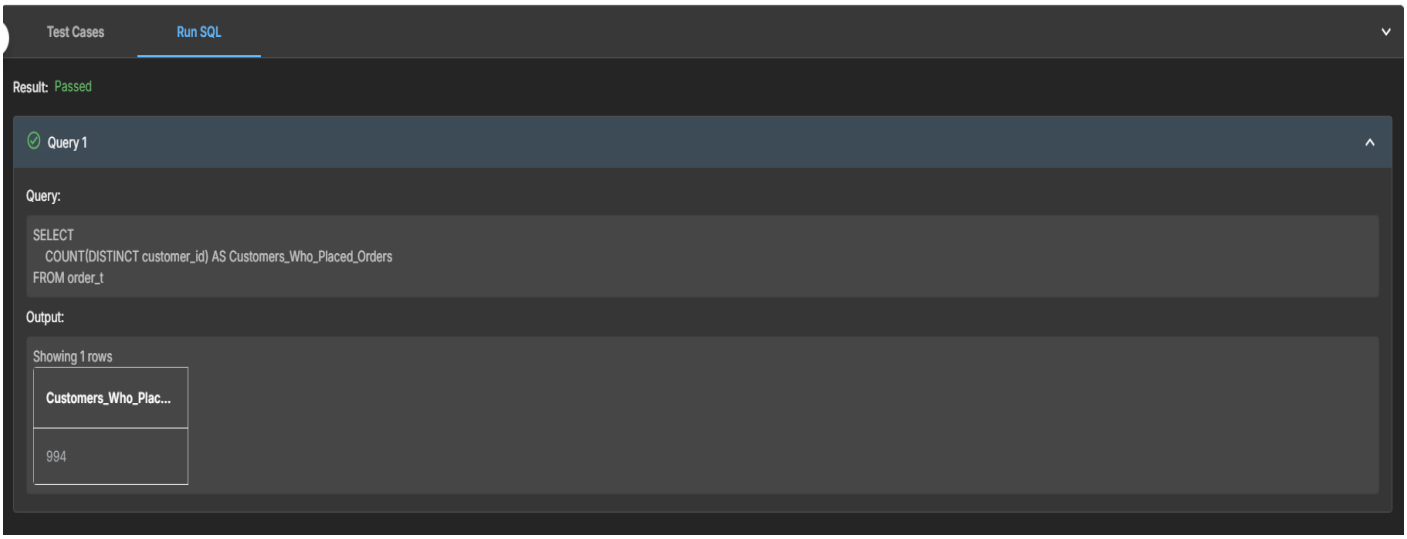
```
SELECT
    COUNT(DISTINCT customer_id) AS Customers_Who_Placed_Orders
FROM order_t;
```

Part 2:

```
SELECT
    c.state,
    COUNT(DISTINCT customer_id) AS num_customers
FROM customer_t c
    JOIN order_t o USING(customer_id)
WHERE quantity IS NOT NULL
GROUP BY
    c.state
ORDER BY
    num_customers DESC;
```

Output:

Part 1:



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Part 2:

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```
SELECT
    c.state,
    COUNT(DISTINCT customer_id) AS num_customers
FROM customer_t c
    JOIN order_t o USING(customer_id)
WHERE quantity IS NOT NULL
GROUP BY
    c.state
ORDER BY
    num_customers DESC
```

Output:

Showing first 10 rows out of 49 rows

state	num_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33

Observations and Insights:

- There is a total of 994 customers who have placed orders.
- Texas, California, and Florida have the greatest number of customers who have placed orders.
- Texas and California both have 97 total customers who have placed orders.
- Ohio and Colorado have fewer customers at 33 each.

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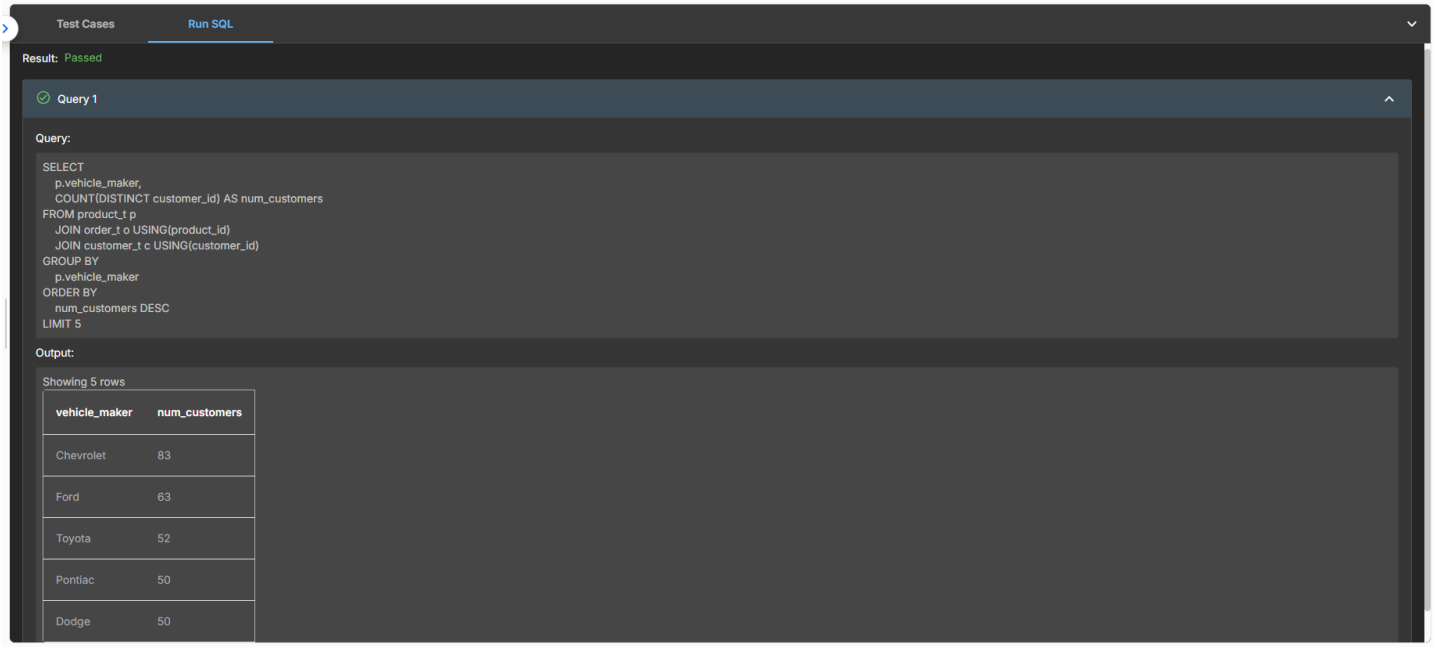
Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT customer_id) AS num_customers
FROM product_t p
    JOIN order_t o USING(product_id)
    JOIN customer_t c USING(customer_id)
GROUP BY
    p.vehicle_maker
ORDER BY
    num_customers DESC
LIMIT 5;
```

Output:



vehicle_maker	num_customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Chevrolet is the top preferred vehicle with 83 customers in the New Wheels database.
- Ford is the second most preferred vehicle with 63 customers.
- Toyota at 52, and Pontiac and Dodge round out the bottom of the top five with 50 customers each.

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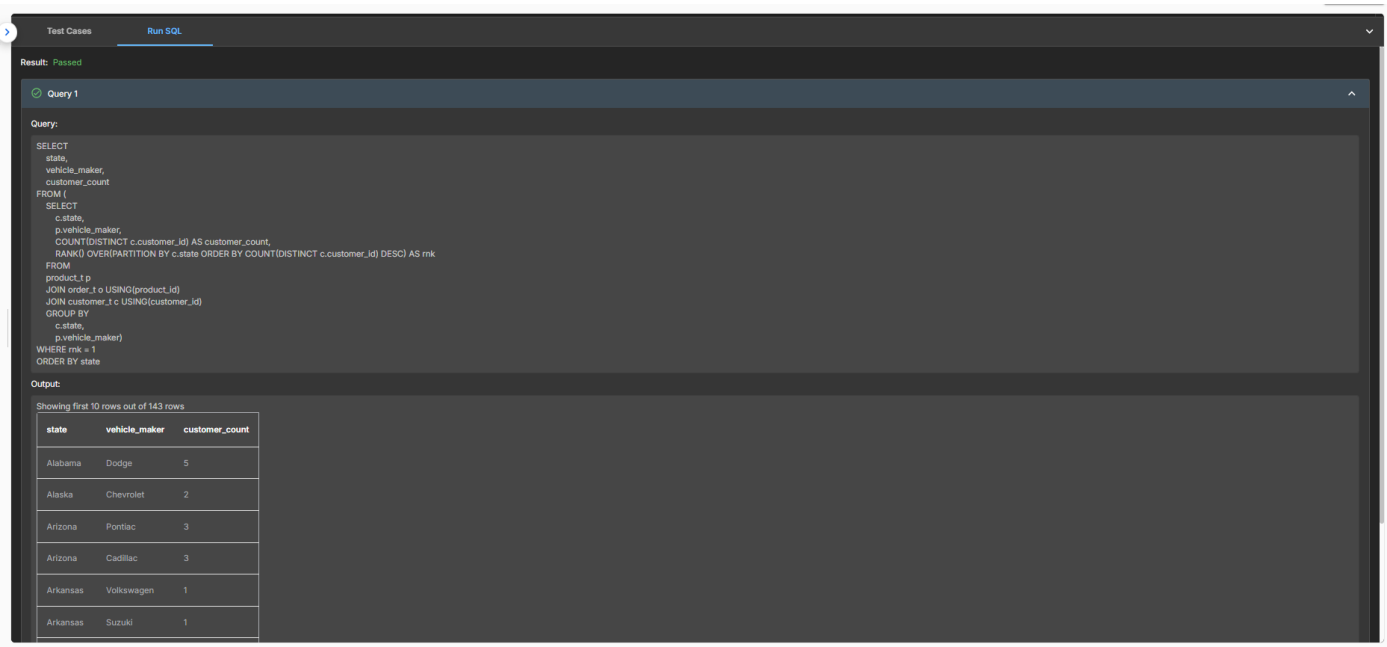
Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
    state,
    vehicle_maker,
    customer_count
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT c.customer_id) AS customer_count,
        RANK() OVER(PARTITION BY c.state ORDER BY COUNT(DISTINCT c.customer_id) DESC) AS rnk
    FROM
    product_t p
    JOIN order_t o USING(product_id)
    JOIN customer_t c USING(customer_id)
    GROUP BY
        c.state,
        p.vehicle_maker)
WHERE rnk = 1
ORDER BY state;
```

Output:



state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1

Observations and Insights:

- Dodge was the most preferred maker in Alabama with a customer count of 5.
- Pontiac and Cadillac were equally preferred in Arizona with a count of 3 each.

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Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

Part 1: Finding the Overall Average Rating given by customers

```
SELECT
    ROUND(AVG(feedback_num_value), 2) as Overall_Avg_Rating
FROM (
    SELECT
        o.quarter_number AS QuarterNumber,
        CASE
            WHEN o.customer_feedback = 'Very Bad' THEN 1
            WHEN o.customer_feedback = 'Bad' THEN 2
            WHEN o.customer_feedback = 'Okay' THEN 3
            WHEN o.customer_feedback = 'Good' THEN 4
            WHEN o.customer_feedback = 'Very Good' THEN 5
            ELSE NULL
        END AS feedback_num_value
    FROM order_t o
) AS Customer_Feedback_Scores;
```

Part 2: The Average Rating in each quarter

```
SELECT
    QuarterNumber,
    ROUND(AVG(feedback_num_value), 2) as Quarterly_Avg_Rating
FROM (
    SELECT
        o.quarter_number AS QuarterNumber,
        CASE
            WHEN o.customer_feedback = 'Very Bad' THEN 1
            WHEN o.customer_feedback = 'Bad' THEN 2
            WHEN o.customer_feedback = 'Okay' THEN 3
            WHEN o.customer_feedback = 'Good' THEN 4
            WHEN o.customer_feedback = 'Very Good' THEN 5
            ELSE NULL
        END AS feedback_num_value
    FROM order_t o
) AS Customer_Feedback_Scores
GROUP BY
    QuarterNumber
ORDER BY
    QuarterNumber;
```

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Question 4, Part 1:

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```
SELECT
  ROUND(AVG(feedback_num_value), 2) as Overall_Avg_Rating
FROM (
  SELECT
    o.quarter_number AS QuarterNumber,
    CASE
      WHEN o.customer_feedback = 'Very Bad' THEN 1
      WHEN o.customer_feedback = 'Bad' THEN 2
      WHEN o.customer_feedback = 'Okay' THEN 3
      WHEN o.customer_feedback = 'Good' THEN 4
      WHEN o.customer_feedback = 'Very Good' THEN 5
      ELSE NULL
    END AS feedback_num_value
  FROM order_t o
) AS Customer_Feedback_Scores
```

Output:

Showing 1 rows

Overall_Avg_Rating
3.14

Question 4, Part 2:

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```
SELECT
  QuarterNumber,
  ROUND(AVG(feedback_num_value), 2) as Quarterly_Avg_Rating
FROM (
  SELECT
    o.quarter_number AS QuarterNumber,
    CASE
      WHEN o.customer_feedback = 'Very Bad' THEN 1
      WHEN o.customer_feedback = 'Bad' THEN 2
      WHEN o.customer_feedback = 'Okay' THEN 3
      WHEN o.customer_feedback = 'Good' THEN 4
      WHEN o.customer_feedback = 'Very Good' THEN 5
      ELSE NULL
    END AS feedback_num_value
  FROM order_t o
) AS Customer_Feedback_Scores

GROUP BY
  QuarterNumber
ORDER BY
  QuarterNumber
```

Output:

Showing 4 rows

QuarterNumber	Quarterly_Avg_Rating
1	3.55
2	3.35
3	2.96
4	2.4

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Question 4 Observations and Insights:

- The overall average rating from customer feedback is 3.14. This returns an overall average of “Okay” when all customer feedback is averaged together.
- The first quarter had the highest overall customer feedback average with 3.55.
- The fourth quarter had the lowest overall customer feedback average with 2.4.
- There appears to be a negative trend of decreasing customer feedback averages over the course of the year.

See next page for Question 5:

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Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    QuarterNumber,
    ROUND(SUM(CASE WHEN feedback_num_value = 1 THEN 1 ELSE 0 END) * 100.0
/COUNT(feedback_num_value), 2) AS 'Very Bad(%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 2 THEN 1 ELSE 0 END) * 100.0
/COUNT(feedback_num_value), 2) AS 'Bad (%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 3 THEN 1 ELSE 0 END) * 100.0
/COUNT(feedback_num_value), 2) AS 'Okay (%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 4 THEN 1 ELSE 0 END) * 100.0
/COUNT(feedback_num_value), 2) AS 'Good (%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 5 THEN 1 ELSE 0 END) * 100.0
/COUNT(feedback_num_value), 2) AS 'Very Good (%)'
FROM (
    SELECT
        o.quarter_number AS QuarterNumber,
        CASE
            WHEN o.customer_feedback = 'Very Bad' THEN 1
            WHEN o.customer_feedback = 'Bad' THEN 2
            WHEN o.customer_feedback = 'Okay' THEN 3
            WHEN o.customer_feedback = 'Good' THEN 4
            WHEN o.customer_feedback = 'Very Good' THEN 5
            ELSE NULL
        END AS feedback_num_value
    FROM order_t o
) AS Customer_Feedback_Scores

GROUP BY
    QuarterNumber
ORDER BY
    QuarterNumber;
```

See Output, Observations, and Insights on the next page:

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Output:

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```
SELECT
    QuarterNumber,
    ROUND(SUM(CASE WHEN feedback_num_value = 1 THEN 1 ELSE 0 END) * 100.0 /COUNT(feedback_num_value), 2) AS 'Very Bad(%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 2 THEN 1 ELSE 0 END) * 100.0 /COUNT(feedback_num_value), 2) AS 'Bad (%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 3 THEN 1 ELSE 0 END) * 100.0 /COUNT(feedback_num_value), 2) AS 'Okay (%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 4 THEN 1 ELSE 0 END) * 100.0 /COUNT(feedback_num_value), 2) AS 'Good (%)',
    ROUND(SUM(CASE WHEN feedback_num_value = 5 THEN 1 ELSE 0 END) * 100.0 /COUNT(feedback_num_value), 2) AS 'Very Good (%)'
FROM (
    SELECT
        o.quarter_number AS QuarterNumber,
        CASE
            WHEN o.customer_feedback = 'Very Bad' THEN 1
            WHEN o.customer_feedback = 'Bad' THEN 2
            WHEN o.customer_feedback = 'Okay' THEN 3
            WHEN o.customer_feedback = 'Good' THEN 4
            WHEN o.customer_feedback = 'Very Good' THEN 5
            ELSE NULL
        END AS feedback_num_value
    FROM order_t o
) AS Customer_Feedback_Scores

GROUP BY
    QuarterNumber
ORDER BY
    QuarterNumber
```

Output:

Showing 4 rows

QuarterNumber	Very Bad(%)	Bad (%)	Okay (%)	Good (%)	Very Good (%)
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- There is an upward trend in Very Bad and Bad customer ratings from quarter 1 through quarter 4.
- We also see a downward trend in Good and Very Good ratings from quarter 1 through quarter 4.
- Overall, customers have become more dissatisfied throughout the year.

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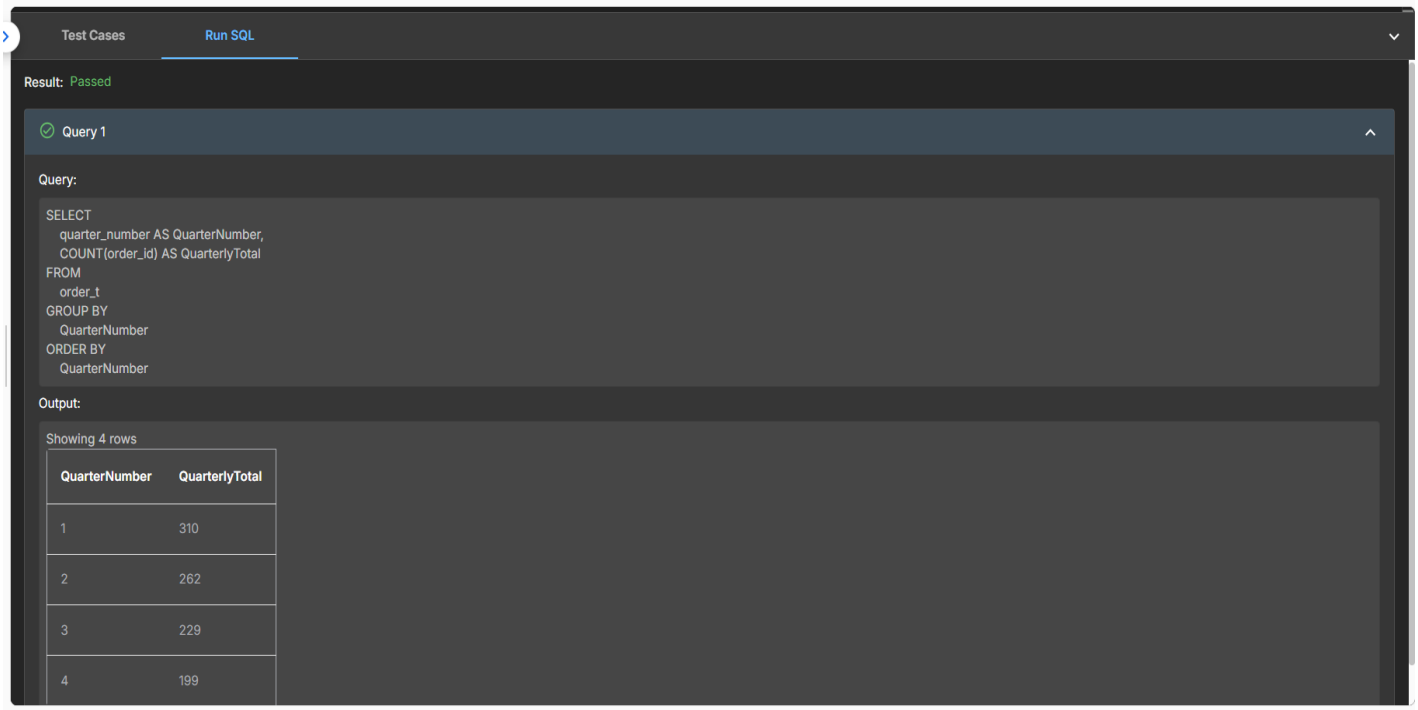
Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    quarter_number AS QuarterNumber,
    COUNT(order_id) AS QuarterlyTotal
FROM
    order_t
GROUP BY
    QuarterNumber
ORDER BY
    QuarterNumber;
```

Output:



QuarterNumber	QuarterlyTotal
1	310
2	262
3	229
4	199

Observations and Insights:

- There is a downward trend in orders from quarter 1 through quarter 4.
- Orders were their highest in quarter 1 with 310 total orders.
- Close to 50 orders were lost between quarter 1 and 2 and a subsequent loss of close to 30 orders between quarters 2 and 3 as well as 3 and 4.

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Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

Question 7, Part 1: Find the net revenue generated by the company

```
SELECT
    ROUND(SUM(quantity * vehicle_price * (1 - (discount / 100))), 2)
        AS Total_Net_Revenue
FROM order_t;
```

Part 2: QoQ % Change in Net Revenue

```
SELECT
    Sales_Quarter,
    ROUND(Net_Revenue, 2) AS Net_Revenue,
    ROUND(LAG(Net_Revenue) OVER(ORDER BY Sales_Quarter), 2) AS Previous_Quarter_Revenue,
    ROUND((Net_Revenue - LAG(Net_Revenue) OVER (ORDER BY Sales_Quarter)) * 100.0
    / LAG(Net_Revenue) OVER (ORDER BY Sales_Quarter),2) AS QoQ_Percentage_Change
FROM (
    SELECT
        quarter_number AS Sales_Quarter,
        SUM(quantity*vehicle_price*(1-(discount / 100))) AS Net_Revenue
    FROM
        order_t
    GROUP BY
        quarter_number
) AS QuarterlyRevenue

ORDER BY
    Sales_Quarter;
```

See Output, Observations, and Insights on the next page:

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Output:

Part 1

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```
SELECT
    ROUND(SUM(quantity * vehicle_price * (1 - (discount / 100))), 2)
        AS Total_Net_Revenue
FROM order_t
```

Output:

Showing 1 rows

Total_Net_Revenue
124714086.32

Part 2:

Test Cases | Run SQL

Result: Passed

Query 1

Query:

```
SELECT
    Sales_Quarter,
    ROUND(Net_Revenue, 2) AS Net_Revenue,
    ROUND(LAG(Net_Revenue) OVER(ORDER BY Sales_Quarter), 2) AS Previous_Quarter_Revenue,
    ROUND((Net_Revenue - LAG(Net_Revenue) OVER (ORDER BY Sales_Quarter)) * 100.0
    / LAG(Net_Revenue) OVER (ORDER BY Sales_Quarter),2) AS QoQ_Percentage_Change
FROM (
    SELECT
        quarter_number AS Sales_Quarter,
        SUM(quantity*vehicle_price*(1-(discount / 100))) AS Net_Revenue
    FROM
        order_t
    GROUP BY
        quarter_number
) AS QuarterlyRevenue

ORDER BY
    Sales_Quarter
```

Output:

Showing 4 rows

Sales_Quarter	Net_Revenue	Previous_Quarter_Rev...	QoQ_Percentage_Cha...
1	39421580.16		
2	32715830.34	39421580.16	-17.01
3	29229896.19	32715830.34	-10.66
4	23346779.63	29229896.19	-20.13

Observations and Insights:

- The total Net Revenue is $124,714,086.32. Net Revenue subtracts any discounts included in sales.
- The percent change was negative for each quarter starting from quarter 1.
- Quarter 2 saw a loss of 17.01 % in Net Revenue from the previous quarter.
- Quarter 3 had the highest Net Revenue loss at -10.66 %.
- Quarter 4 lost -20.13 % from the previous quarter.

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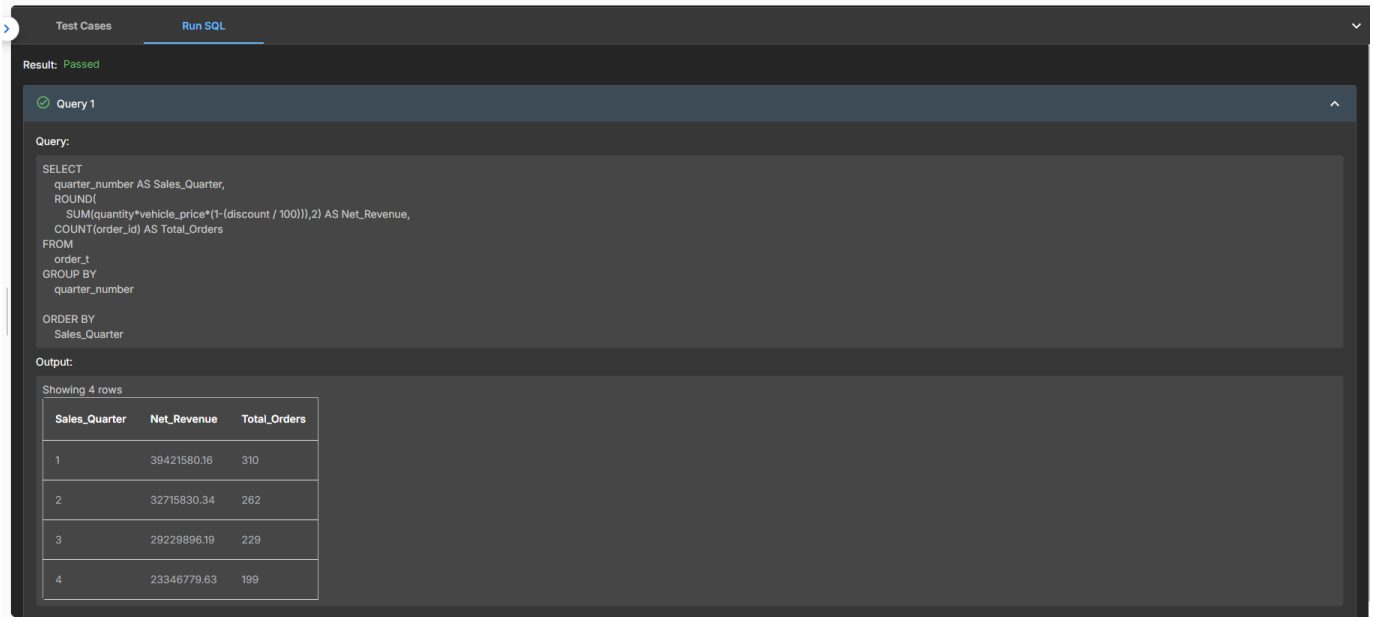
Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number AS Sales_Quarter,
    ROUND(
        SUM(quantity*vehicle_price*(1-(discount / 100))),2) AS Net_Revenue,
    COUNT(order_id) AS Total_Orders
FROM
    order_t
GROUP BY
    quarter_number

ORDER BY
    Sales_Quarter;
```

Output:



Sales_Quarter	Net_Revenue	Total_Orders
1	39421580.16	310
2	32715830.34	262
3	29229896.19	229
4	23346779.63	199

Observations and Insights:

- There is an overall negative trajectory, or downward trend for Net Revenue and Total Orders for each quarter.
- There are losses of roughly $7 million between quarter 1 and quarter 2, $3 million between quarters 2 and 3, and $6 million between quarters 3 and 4.
- Total Orders have dropped from 310 in the first quarter all the way to 199 in the last quarter.

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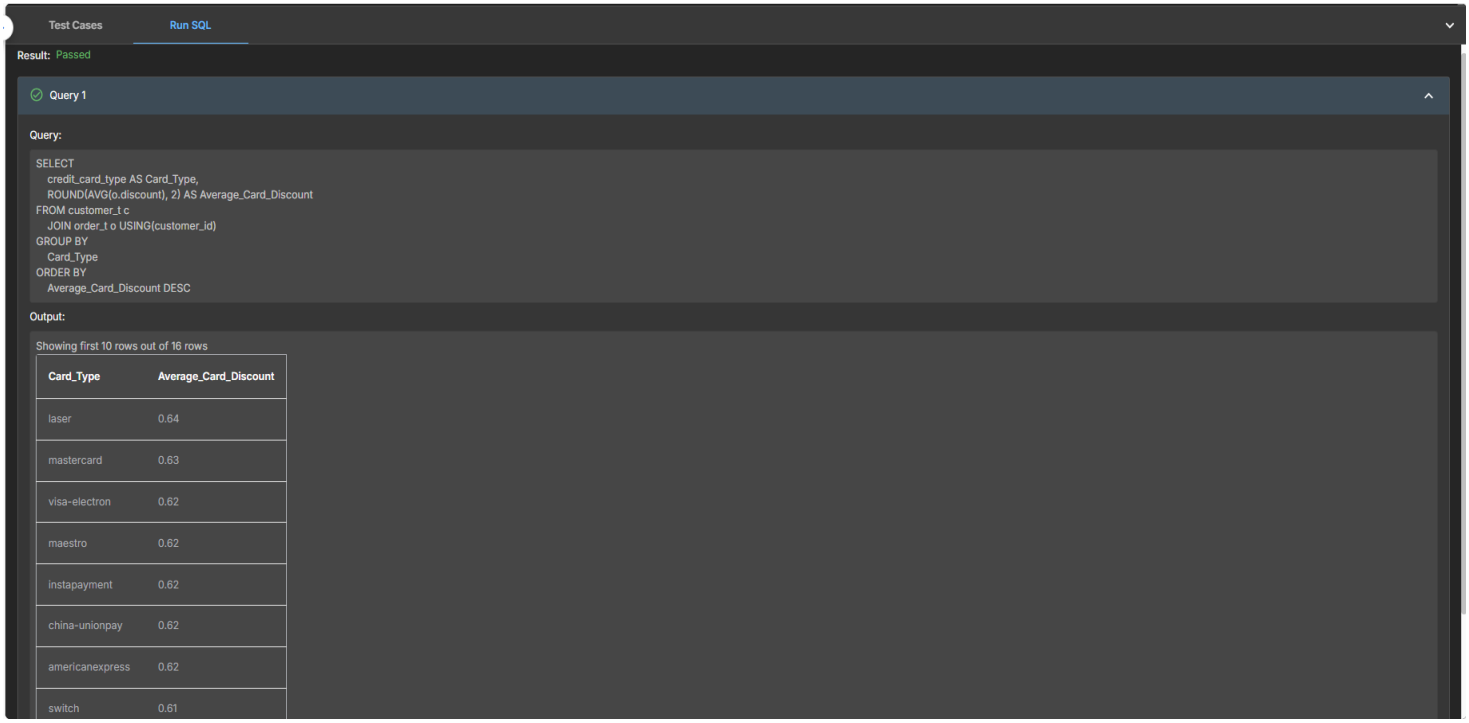
Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    credit_card_type AS Card_Type,
    ROUND(AVG(o.discount), 2) AS Average_Card_Discount
FROM customer_t c
    JOIN order_t o USING(customer_id)
GROUP BY
    Card_Type
ORDER BY
    Average_Card_Discount DESC;
```

Note: The '`discount`' field in the '`order_t`' table is stored as a percentage, i.e. `0.6` represents a discount of `0.6%`, not `60%`.

Output:



Card_Type	Average_Card_Discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
switch	0.61

Observations and Insights:

- Laser cardholders have the largest average discount at 0.64%.
- Mastercard users receive a discount of 0.63%.
- Visa-electron, Maestro, Instapayment, and China-Union Pay all receive a discount of 0.62%.

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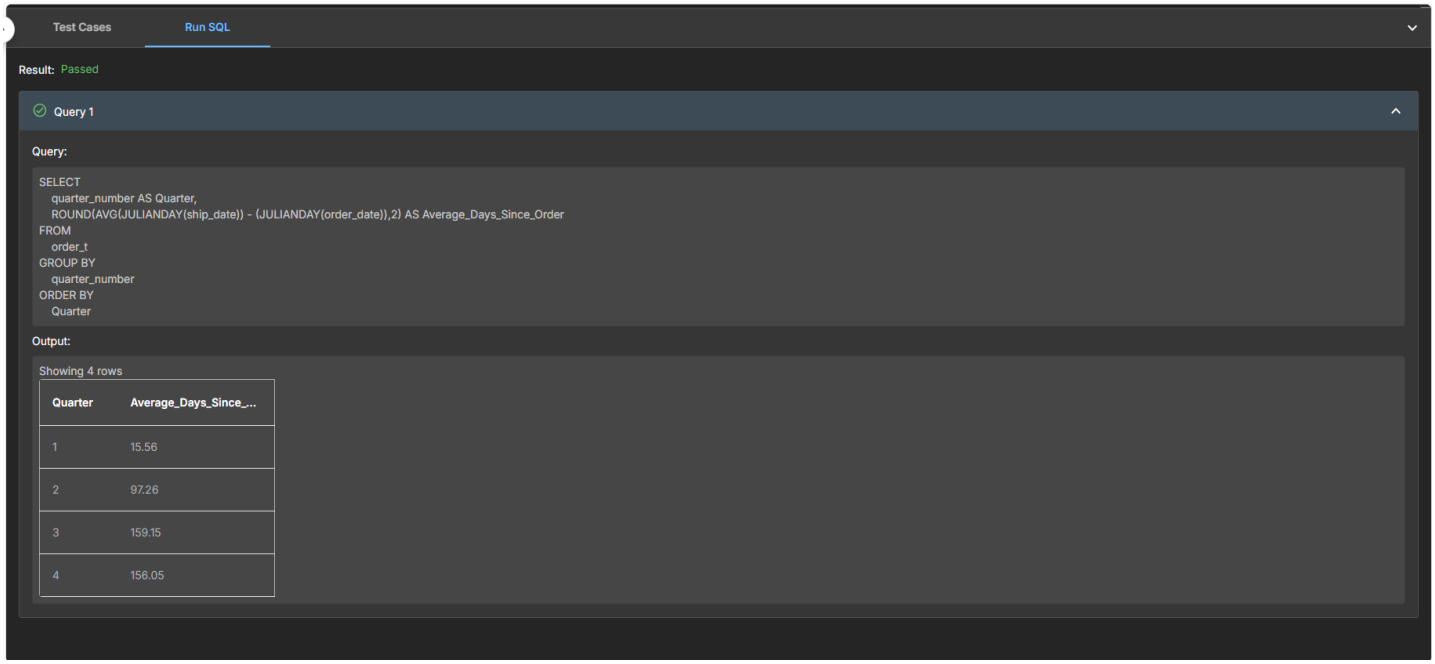
Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number AS Quarter,
    ROUND(AVG(JULIANDAY(ship_date)) - (JULIANDAY(order_date)),2) AS Average_Days_Since_Order
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    Quarter;
```

Output:



Quarter	Average_Days_Since_...
1	15.56
2	97.26
3	159.15
4	156.05

Observations and Insights:

- The average days between the order date and ship date have gotten progressively longer.
- Quarter 1 had the shortest average time taken to ship orders at just over 2 weeks.
- Quarter 3 and Quarter 4 posted average days to ship from 160 days to 156 days respectively.

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Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
$125,482,804.43	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
$23,496,008.22	199	98	44.1

Note: These values must be derived using SQL queries. See Appendix for queries. Some of them may have already been obtained while answering previous questions.

Note: *% Good Feedback is calculated to include both Good and Very Good Ratings from customers.*

Business Recommendations

- Average shipping time was approximately 15 days for Q1, but increased significantly in later quarters. New Wheels should conduct an internal audit to check warehouse staffing levels, inventory, as well as shipper performance to see why delays are happening.

- The rise in shipping delays coincides with a decline in customer satisfaction. Customers give more neutral, to negative feedback when delivery times are longer. An improvement in shipping times is likely to have a positive effect on customer sentiment.

- New Wheels should prioritize operational movement in the top three states with the most customers and focusing on reducing shipping times between the order date and the ship date.

- Aim for orders to be delivered in 15 days or less. Achieving this goal will enhance customer experience for the largest customer base. Targeting these high-traffic states could also increase customer retention and lead to more positive feedback.

See Appendix for queries pertaining to the Business Matrix.

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Appendix

Business Matrix Derived Queries:

Total Revenue:

```
SELECT
    ROUND(SUM(quantity*vehicle_price),2) AS Total_Revenue
FROM
    order_t;
```

Total Orders

```
SELECT
    COUNT(order_id) AS Total_Orders
FROM
    order_t;
```

Total Customers

```
SELECT

    COUNT(DISTINCT customer_id) AS Customers_Who_Placed_Orders

FROM order_t;
```

Average Rating

```
SELECT
    ROUND(AVG(feedback_num_value), 2) as Overall_Avg_Rating
FROM (
    SELECT
        o.quarter_number AS QuarterNumber,
        CASE
            WHEN o.customer_feedback = 'Very Bad' THEN 1
            WHEN o.customer_feedback = 'Bad' THEN 2
            WHEN o.customer_feedback = 'Okay' THEN 3
            WHEN o.customer_feedback = 'Good' THEN 4
            WHEN o.customer_feedback = 'Very Good' THEN 5
            ELSE NULL
        END AS feedback_num_value
    FROM order_t o
) AS Customer_Feedback_Scores;
```

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Last Quarter Revenue:

Gross Revenue: (excluding discounts or deductions)

```
SELECT
    quarter_number AS Quarter_Number,
    ROUND(SUM(quantity*vehicle_price),2) AS Total_Revenue
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    Quarter_Number DESC
LIMIT 1;
```

Last Quarter Orders:

```
SELECT
    quarter_number AS Quarter_Number,
    COUNT(order_id) AS Total_Orders
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    Quarter_Number DESC
LIMIT 1;
```

Average Days to Ship:

```
SELECT
    ROUND(
        AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)),
        0
    ) AS Average_Days_Since_Order
FROM
    order_t;
```

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% Good Feedback:

(including "Good" and "Very Good" customer ratings)

```
SELECT
ROUND(
    (SUM(CASE WHEN feedback_num_value = 4 THEN 1 ELSE 0 END) +
     SUM(CASE WHEN feedback_num_value = 5 THEN 1 ELSE 0 END))
    * 100.0 / COUNT(feedback_num_value),
    2
) AS 'Good + Very Good (%)'
FROM (
    SELECT
        o.quarter_number AS QuarterNumber,
        CASE
            WHEN o.customer_feedback = 'Very Bad' THEN 1
            WHEN o.customer_feedback = 'Bad' THEN 2
            WHEN o.customer_feedback = 'Okay' THEN 3
            WHEN o.customer_feedback = 'Good' THEN 4
            WHEN o.customer_feedback = 'Very Good' THEN 5
            ELSE NULL
        END AS feedback_num_value
    FROM order_t o
) AS Customer_Feedback_Scores;
```

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